ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** biosphere	Sphere of life.
2	**(c)** fresh snow	High albedo.
3	**(a)** stratosphere	About 15 to 35 km.
4	**(a)** water (hydrological) cycle	Cycle of water.
5	**(a)** geosphere	Rocky part.
6	**(a)** ocean water or forests	Low albedo.
7	**(c)** harmful ultraviolet radiation	Absorbs UV.
8	**(c)** cycling of nutrients through living and non-living parts of the Earth	Matter recycles.
9	**(c)** hydrosphere	Water.
10	**(c)** sunlight falls obliquely and is spread over a larger area	Less heating.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Geosphere: rocks and soil; hydrosphere: water (1); Atmosphere: gases; biosphere: life (1).	1+1
12	Cyclic movement of nutrients through living and non-living parts (1); Examples: carbon and nitrogen cycles (1).	1+1
13	Provides gases like oxygen and carbon dioxide (1); Protects from UV rays and keeps the Earth warm (1).	1+1
14	Energy from the Sun reaching the Earth (1); Drives climate, water cycle and photosynthesis (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Evaporation from water bodies (hydrosphere) (1); Condensation and rain from the atmosphere (1); Rain affects soil (geosphere) and plants (biosphere) (1).	1+1+1
16	At the equator rays are nearly vertical (1); At the poles they are slanting (1); Spread over a larger area, less heating (1).	1+1+1
17	Convert nitrogen into nitrates (1); Enrich the soil (1); Reduce the need for fertilisers (1).	1+1+1
	OR: Gases trap heat, keeping the Earth warm (1); Burning fuels and deforestation add more CO2 (1); Enhanced effect causes global warming (1).	1+1+1
18	Snow reflects most (1); Forests and oceans absorb (1); Melting ice lowers albedo, increasing warming (1).	1+1+1
	OR: Absorb CO2 (1); Store carbon in wood (1); Regulate local climate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Four spheres (1); Energy from the Sun (1); Matter cycles (1); Interactions (1); Human influence (1).	1+1+1+1+1
	OR: Fixation by bacteria and lightning (1); Absorption of nitrates (1); Passing through the food chain (1); Decay by decomposers (1); Return to the air (1).	1+1+1+1+1
20	Light takes 8 minutes (1); More at the equator (1); Part reflected by clouds and surface (1); Part absorbed (1); Re-radiated as heat (1).	1+1+1+1+1
	OR: Albedo definition (1); Snow high, forest low (1); Melting ice reduces albedo (1); More warming (feedback) (1); Urban surfaces absorb (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.

- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs $\mathrm{CO_2}$ and releases $\mathrm{O_2}$ (plants release $\mathrm{CO_2}$ only in respiration).
- Not converting units in rainwater problems: $\mathrm{area} \times \mathrm{depth}$ gives $\mathrm{m^3}$, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** chlorofluorocarbons (CFCs)	Used in refrigerators and sprays.
2	**(c)** atmosphere	78 per cent of air.
3	**(a)** atmosphere	Air.
4	**(c)** the Sun's rays fall almost vertically	Direct rays.
5	**(a)** Montreal Protocol	1987.
6	**(b)** nitrogen-fixing bacteria	Rhizobium.
7	**(b)** biosphere	Life.
8	**(c)** greenhouse gases	e.g. CO2.
9	**(a)** Antarctica	Polar region.
10	**(a)** transpiration	Through stomata.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Using public transport or cycling (1); Saving electricity and planting trees (1).	1+1
12	Fraction of sunlight reflected by a surface (1); Snow or ice (1).	1+1
13	Sunlight falls obliquely and spreads over a larger area (1); More of it is reflected by snow and ice (1).	1+1
14	Harmful: skin cancer, cataract (1); Useful: vitamin D formation, disinfection (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	CFCs release chlorine in the stratosphere (1); More UV reaches the Earth (1); Skin cancer, cataract, damage to crops (1).	1+1+1
16	Melting of glaciers and rise in sea level (1); Changes in rainfall patterns (1); More frequent extreme weather (1).	1+1+1
17	Burning fossil fuels (1); Use of chemical fertilisers (1); Deforestation (1).	1+1+1
	OR: Plants absorb CO2 in photosynthesis (1); Respiration and decay return CO2 (1); Burning fuels adds CO2 (1).	1+1+1
18	Chlorofluorocarbons used in refrigerators and sprays (1); Damage the ozone layer (1); Banned under the Montreal Protocol (1).	1+1+1
	OR: t = d/v (1); = 1.5 x 10^11 / 3 x 10^8 (1); = 500 s, about 8.3 minutes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CO2, methane, nitrous oxide (1); Sources (1); Warming (1); Consequences (1); Mitigation (1).	1+1+1+1+1
	OR: High sunshine (1); Solar cells (1); Solar water heaters (1); Advantages: clean, renewable (1); Limitations: cost, weather (1).	1+1+1+1+1
20	Location (1); UV absorption (1); CFCs (1); Effects (1); Montreal Protocol (1).	1+1+1+1+1
	OR: Mission LiFE meaning (1); Save energy (1); Save water (1); Reduce waste (1); Plant trees, spread awareness (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs CO_2 and releases O_2 (plants release CO_2 only in respiration).
- Not converting units in rainwater problems: $\text{area} \times \text{depth}$ gives m^3, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the Sun	Solar radiation.
2	**(d)** necessary for life on Earth	Keeps Earth warm.
3	**(d)** skin cancer and cataract	Damage.
4	**(a)** plants and animals	Carbon compounds.
5	**(b)** 3 x 10^8 m/s	c.
6	**(c)** carbon dioxide	CO2.
7	**(a)** helps make vitamin D and kills germs	Benefits.
8	**(a)** photosynthesis	Plants.
9	**(c)** 8 minutes	About 150 million km.
10	**(c)** global warming	Climate change.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Trapping of heat radiated by Earth by gases (1); Keeps the Earth warm enough for life (1).	1+1
12	Layer of ozone gas in the stratosphere (1); Absorbs harmful ultraviolet rays (1).	1+1
13	Located in the tropics with high sunshine (1); Many clear sunny days in a year (1).	1+1
14	Evaporation and transpiration (1); Condensation and precipitation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Bacteria fix atmospheric nitrogen (1); Plants absorb nitrates and make proteins (1); Decomposers return nitrogen to the soil and the air (1).	1+1+1
16	Lifestyle for Environment, a mass movement (1); Save energy and water (1); Reduce waste and avoid single-use plastics (1).	1+1+1
17	Clouds reflect part (1); Gases absorb UV and some infrared (1); Only a part reaches the surface (1).	1+1+1
	OR: Sun's heat evaporates water (1); Vapour condenses to clouds (1); Precipitation returns water (1).	1+1+1
18	Photosynthesis (1); Water cycle and winds (1); Heating the Earth (1).	1+1+1
	OR: Using renewable energy (1); Rainwater harvesting (1); Reduce, reuse and recycle (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Evaporation and transpiration (1); Condensation (1); Precipitation (1); Run-off and infiltration (1); Importance (1).	1+1+1+1+1
	OR: Weathering links geosphere with atmosphere and water (1); Photosynthesis links biosphere with air and water (1); Rain formation (1); Cycles of matter (1); Changes in one affect others (1).	1+1+1+1+1
20	Atmosphere reservoir (1); Photosynthesis (1); Respiration and decay (1); Fossil fuels (1); Ocean uptake and human impact (1).	1+1+1+1+1
	OR: Rising CO2 (1); Melting ice (1); Ozone hole (1); Changes in climate (1); Measures to restore (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs CO_2 and releases O_2 (plants release CO_2 only in respiration).
- Not converting units in rainwater problems: $\text{area} \times \text{depth}$ gives m^3, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

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